

II-JF – Cube27 3-Primary Incidence Bockstein/Linking Form and Integral Spectral-Lattice Extension, Cuboctahedral–Stella/FCC–HCP Layer-Stacking Contact-Groupoid Audit, Golden-Hodge Stabilizer Action-Isomorphism and Periodic H^2 /Extension Refinement, PLR Antipodal Quotient/Fusion Scheme and Q -/ P -Polynomial Obstruction Audit, Pisano–CBF4 Relative Period/Residue Pairing and Epstein Functional-Cohomology Descent, plus R35-to-R36 Kepler Forward Descent, Native Lean Replay Gate, and Five-Branch Payload Audit

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17 August 2026

Abstract

II-JF freezes II-JE and keeps the original geometric carrier in first position: the 27 fixed barycentric centers split as the cuboctahedral $E_{13} = L_0 \sqcup L_2$ and resolved-stella $O_{14} = L_1 \sqcup L_3$ sheets. The integral cross-incidence map has Smith torsion $\mathbb{Z}/3$; II-JF identifies an explicit Bockstein generator, a nonzero dual cross-linking value $1/3$, and proves that the independently occurring integral spectral-lattice gluing defect is purely 2-primary. The FCC/HCP three-layer stacking windows form two invariant action-groupoid components. Two supplied classical papers of Sullivan and Epstein–Vogt are integrated only as codimension-three support contexts: they motivate obstruction/hierarchy and stratified spinning/doubling methods, not identities with project carriers. Golden-Hodge stabilizer actions reduce to four module types and their extension-control H^2 groups are computed; all actual stabilizer extensions remain split. The PLR antipode yields a rank-7 quotient association scheme, but both P - and Q -polynomial shortcuts are refuted. The relative orientation residue is exact in the divisor complex and orthogonal to the constant Appell period, while the completed Epstein functional equation descends through $(s - 1/2)^2$. R35→R36 is replayed byte-exactly; the supplied Lean source/.`olean` pair is hash-locked, while native proof-assistant reload remains separated without a compatible runtime. No synthetic object is promoted to one of the 37 active project-value gates.

1 Entry discipline and the primary geometric carrier

II-JE is frozen at **PASS**. The two ledgers remain append-only, truth and mergeability are orthogonal, and the current phase starts from the exact fixed-center decomposition

$$G = \sqrt{2}\{-1, 0, 1\}^3 = E_{13} \sqcup O_{14}, \quad E_{13} = L_0 \sqcup L_2, \quad O_{14} = L_1 \sqcup L_3.$$

The Kepler branch, the Golden-Hodge branch, PLR, and the analytic Appell/Epstein branch are typed descendants or comparison carriers. None replaces this Cube27/cuboctahedral/stella head.

2 Cube27: the 3-primary incidence class

Let

$$B : \mathbb{Z}_O^{14} \longrightarrow \mathbb{Z}_E^{13}, \quad B_{e,o} = 1 \iff \|e - o\|^2 = 1.$$

II-JE proved

$$\text{SNF}(B) = \text{diag}(1^9, 3, 0^3).$$

Hence

$$\ker B \simeq \mathbb{Z}^4, \quad \text{coker } B \simeq \mathbb{Z}^3 \oplus \mathbb{Z}/3.$$

II-JF now fixes the orientation of the torsion class explicitly. Write c for the central E -basis vector and order the 14 odd centers lexicographically as in the verifier. For

$$v = (1, 1, -1, 1, 0, -1, -1, 0, 0, 1, 0, 0, 0, 0)^T$$

one has

$$\boxed{Bv = -3c}, \quad \sum_i v_i = 1.$$

Thus $\bar{v} = v \bmod 3$ is the extra mod-3 cycle and the Bockstein connecting morphism sends it to the torsion generator $[-c]$.

There is a canonical dual torsion witness after the declared sheet orientation. Put

$$\ell_E = -\delta_c + \sum_{e \in L_2} \delta_e, \quad \mathbf{1}_O = (1, \dots, 1)^T.$$

Then

$$\boxed{B^T \ell_E = 3\mathbf{1}_O}.$$

The resulting dual-carrier linking pairing is therefore

$$\boxed{\lambda([-c], [\mathbf{1}_O]) = \frac{1}{3} \mathbf{1}_O^T v = \frac{1}{3} \pmod{\mathbb{Z}}.}$$

HT+NT guard. This is a cross-linking Tor coker $B \times \text{Tor coker } B^T \rightarrow \mathbb{Q}/\mathbb{Z}$. No canonical self-duality of the two torsion carriers has been constructed, so a self-linking interpretation is **SEPARATED**.

3 Integral spectral lattices: a primary-separation theorem

Let A_E, A_O be the two endpoint adjacency matrices. The intertwining relation

$$A_E B = B A_O$$

preserves the rational eigenvalues 6, 2, 0, -2. For each λ let

$$L_{E,\lambda} = \ker(A_E - \lambda I) \cap \mathbb{Z}^{13}, \quad L_{O,\lambda} = \ker(A_O - \lambda I) \cap \mathbb{Z}^{14}.$$

The direct sums of these saturated eigenlattices have indices

$$[\mathbb{Z}^{13} : \bigoplus_{\lambda} L_{E,\lambda}] = 2^{14}, \quad [\mathbb{Z}^{14} : \bigoplus_{\lambda} L_{O,\lambda}] = 2^{16}.$$

Their Smith invariants are

$$E : \quad 1^6, 2^4, 8^2, 16, \quad O : \quad 1^6, 2^4, 4^2, 8, 32.$$

Thus the failure of integral spectral splitting is purely 2-primary. The restrictions of B between the saturated eigenlattices have nonzero Smith invariants

λ	6	2	0	-2
SNF _{$\neq 0$} (B_λ)	(6)	(2, 2, 8)	(1, 1, 1)	(1, 1, 1).

After inverting 2, the unique 3-primary defect is therefore carried by the $\lambda = 6$ block.

Theorem 1 (Primary separation). *The $\mathbb{Z}/3$ incidence Bockstein defect and the integral spectral-lattice gluing defect do not have the same primary source. The former is 3-primary; the latter is 2-primary. The direct claim that the $\mathbb{Z}/3$ class is caused by the spectral-lattice index is **REFUTED**.*

4 FCC/HCP layer stacking and the codimension-three support layer

Use layer labels A, B, C and the local three-layer window carrier

$$\mathcal{W} = \{(a, b, c) \in \{A, B, C\}^3 : a \neq b, c \neq b\}.$$

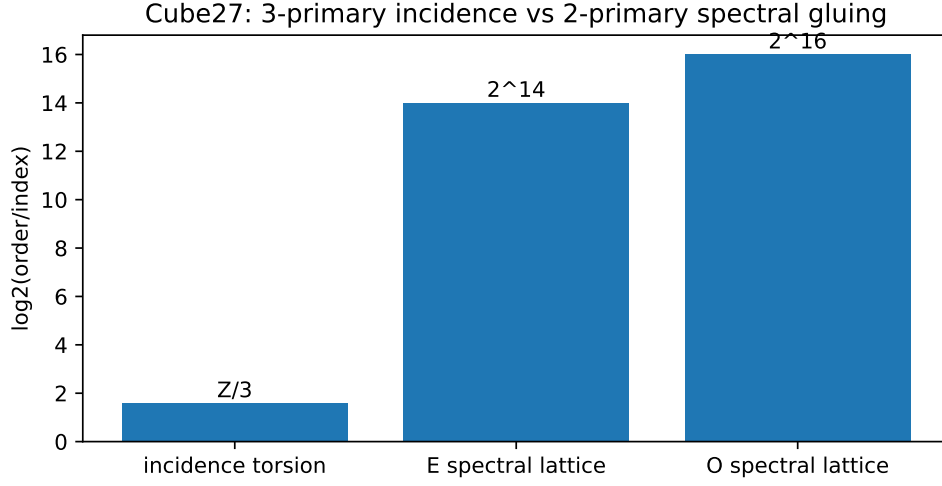


Figure 1: Two independent integral defects: the 3-primary incidence torsion and the 2-primary index of the endpoint spectral lattices.

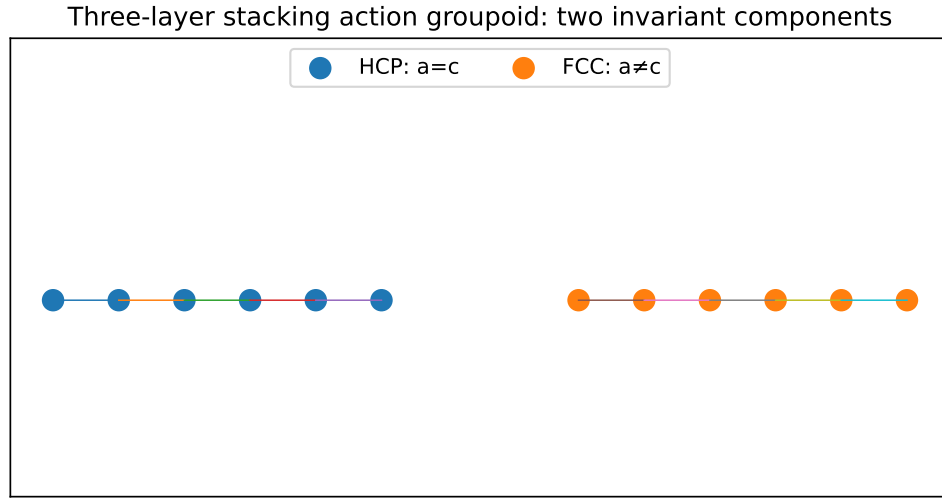


Figure 2: The layer-stacking action groupoid has two invariant components; label symmetries and reversal do not connect FCC to HCP.

It has 12 objects. The action of S_3 on layer labels, optionally extended by stack reversal $(a, b, c) \mapsto (c, b, a)$, preserves the predicate $a = c$. Hence

$$\mathcal{W} = \mathcal{W}_{\text{HCP}} \sqcup \mathcal{W}_{\text{FCC}}, \quad |\mathcal{W}_{\text{HCP}}| = |\mathcal{W}_{\text{FCC}}| = 6.$$

The FCC component maps to the inherited cuboctahedral D_3 kissing shell, so E_{13} is exactly “center + FCC shell”. The HCP component maps to the distinct HCP twelve-neighbour shell. The resolved stella O_{14} is not the HCP shell: their raw carrier cardinalities and contact structures already differ.

Two classical sources supplied with II-JF are archived and hash-locked as *support contexts*. Sullivan’s 1976 construction gives a compact five-dimensional periodic-flow counterexample and explicitly isolates codimension three as a further difficult boundary, discussing obstructions to simple manifold bad sets. Epstein–Vogt subsequently construct a codimension-three example on a compact four-manifold as a polynomial regular level set in $\mathbb{R}^7$; their bad set is a stratified “necklace” of four S^3 pieces alternating with four T^2 pieces, adjacent pieces sharing a circle. Their spinning/doubling construction is therefore particularly relevant as a methodological precedent for stratified/groupoid gluing.

The exact finite shadow used here is only

$$N_8 \simeq C_8, \quad S^3, T^2, S^3, T^2, \dots$$

and rotation by two edges gives a C_4 action on each four-element bead-type orbit.

HT+NT guard. The Sullivan and Epstein–Vogt objects are not Cube27, PLR, Pisano/CBF4 or Kepler carriers. Their role is documentary and methodological: codimension-three obstruction, hierarchy, stratification, spinning and doubling. Any raw identification is **SEPARATED/PROVEN-NOT-MERGEABLE**.

5 Golden-Hodge: action types and extension-control H^2

The 384-element Frobenius-symplectic centralizer acts on the 64 quadratic refinements. II-JE resolved seven orbits. II-JF compares the elementary-abelian kernel actions rather than only their orbit size and Arf invariant.

The seven actions collapse to four $\mathbb{F}_2[Q]$ -module types:

1. the two size-4 orbits (Arf 0 and 1) have isomorphic $K \simeq \mathbb{F}_2^4$ actions with $Q \simeq S_3$;
2. the size-8 orbit has $K \simeq \mathbb{F}_2^3$, $Q \simeq S_3$;
3. the two size-12 rank-2 involution actions pair across Arf;
4. the two size-12 rank-1 involution actions pair across Arf.

The rank-1 and rank-2 C_2 actions are nonisomorphic.

For $Q = C_2$ in characteristic two,

$$H^2(C_2; K) \simeq K^{C_2}/(g-1)K.$$

For the S_3 cases, the normal C_3 has odd order, so extension control reduces to the C_2 action on K^{C_3} . In the canonical orbit ordering the dimensions are

$$\boxed{2, 2, 1, 0, 2, 0, 2.}$$

Despite these nonzero spaces of possible extensions, every actual stabilizer projection used here has an explicit complement. Therefore its factor-set class is

$$\boxed{[\kappa_{\text{ext}}] = 0 \in H^2(Q; K)} \quad (\text{ZERO}).$$

HT+NT guard. $H^2(Q; K)$ controls these finite group extensions. It is neither the full $H^2(\text{Stab}; \mathbb{F}_4)$ nor the integral Golden-Hodge/Yoneda/Tate derived carrier.

6 PLR: antipodal quotient and the failure of metric/cometric shortcuts

II-JE found the unique nonidentity valency-one relation A_{ant} on the 320 PLR objects. Multiplication by A_{ant} pairs the rank-14 relations as

$$(0, 1), (2, 3), (4, 5), (6, 7), (8, 9), (10, 11), (12, 13).$$

The 320 vertices themselves form 160 antipodal pairs. Quotienting by this canonical involution produces a symmetric rank-7 association scheme with valencies

$$\boxed{1, 6, 9, 18, 18, 36, 72}$$

and primitive multiplicities

$$\boxed{1, 9, 15, 15, 30, 45, 45.}$$

Its exact first eigenmatrix is

$$P = \begin{pmatrix} 1 & 6 & 9 & 18 & 18 & 36 & 72 \\ 1 & 6 & 9 & -2 & -2 & -4 & -8 \\ 1 & -2 & 1 & -6 & 10 & 4 & -8 \\ 1 & 2 & -3 & 6 & 6 & -12 & 0 \\ 1 & -2 & 1 & 6 & -2 & 4 & -8 \\ 1 & -2 & 1 & -2 & -2 & -4 & 8 \\ 1 & 2 & -3 & -2 & -2 & 4 & 0 \end{pmatrix}, \quad PQ = 160I_7.$$

For the six nonidentity adjacency columns the numbers of distinct eigenvalues are

$$\boxed{3, 3, 4, 4, 4, 4},$$

so no relation can generate a diameter-6 P -polynomial ordering. For the six nontrivial columns of Q the distinct-value counts are

$$\boxed{2, 5, 4, 5, 4, 6},$$

so no Q -polynomial ordering exists either.

The canonical antipode therefore rigidifies an intrinsic C_2 quotient but still does not choose one of the four oriented generators of a C_{12} frame.

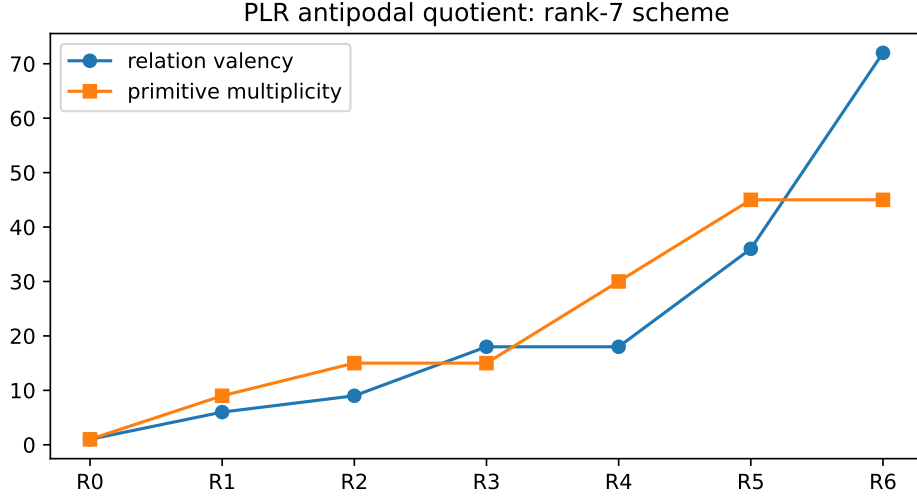


Figure 3: Valencies and primitive multiplicities of the antipodal PLR quotient.

7 Relative residue, Appell period and Epstein descent

Cellulate the ellipse/circle by the four divisor points $D = \{xy = 0\}$ and orient the four arcs cyclically. Then

$$H^1(E, D; \mathbb{Z}) \simeq \mathbb{Z}^4.$$

The orientation residue

$$\alpha = (1, -1, 1, -1)$$

is exact in the relative divisor presentation. Indeed for

$$r = (0, 1, 0, 1) \in C^0(D; \mathbb{Z})$$

one has

$$\boxed{\delta r = \alpha}.$$

The map from relative to absolute degree one sums the four arc coefficients, hence α has zero absolute period.

The constant Appell quarter-period cochain is

$$p = \frac{\pi}{4}(1, 1, 1, 1), \quad \sum p_i = \pi,$$

and its pairing with the alternating residue is

$$\boxed{\langle p, \alpha \rangle = \frac{\pi}{4}(1 - 1 + 1 - 1) = 0} \quad (\text{ZERO}).$$

For the completed Epstein object Λ_5 , the functional equation

$$\Lambda_5(s) = \Lambda_5(1 - s)$$

becomes evenness in the centered coordinate $\sigma = s - \frac{1}{2}$. Hence the invariant analytic carrier descends through

$$\boxed{u = \sigma^2 = (s - \frac{1}{2})^2}.$$

The relative class α is a sign line under quarter-turn, whereas the completed Epstein carrier is invariant under reflection. Their direct identification is therefore **REFUTED/PROVEN-NOT-MERGEABLE**.

8 Kepler R35→R36 and the proof-assistant gate

The supplied R36 lineage is replayed against the supplied R35 archive. The parent bundle hash and all declared input anchors agree byte-for-byte. The decision state changes as follows:

$$\boxed{737 \rightarrow 737 \text{ classes}, \quad 8711 \rightarrow 8871 \text{ boxes},}$$

948 $\rightarrow$ 901 exact S_4 orbits, 1348 $\rightarrow$ 1298 subleaves,

7902 $\rightarrow$ 7736 active cells, 6 $\rightarrow$ 6 open routes.

Thus R36 closes 47 endpoint orbits and 50 subleaves, with one partially reduced orbit and no transverse class closure. The R36 local Putinar packet passes its exact rational identity and interval replay; its certified lower bound is strictly positive. The provided core translation source and `.olean`, together with the embedded R36 Lean source, are hash-locked. The binary surface contains the version marker `Lean 4.30.0-pre` and the expected exported symbols. No compatible executable is present in the current workspace, so native reload/recompile remains **SEPARATED**; byte provenance is not silently upgraded to theorem-assistant replay.

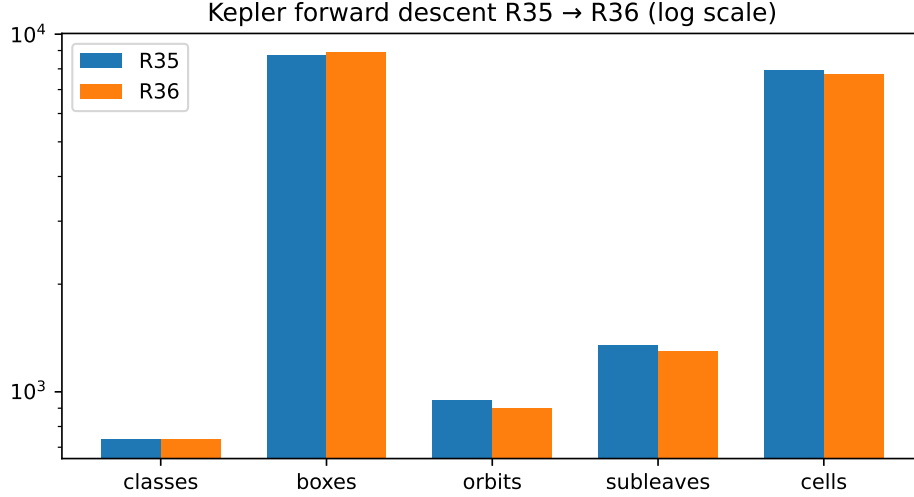


Figure 4: R35 $\rightarrow$ R36 forward descent. The box count rises because of refinement while endpoint orbit/subleaf/cell counts decrease.

9 Five-branch payload audit and append-only checkpoint

The inherited ABI was rescanned for Fibonacci, Stone, Bornological, Rees and H_3 . No new source-bound actual project packet was supplied. The finite/synthetic lanes remain useful test objects but do not close a project value. The 37 active project-specific NT gates are unchanged.

The append-only checkpoint is

PASS = 699, ZERO = 30, SEPARATED = 193, REFUTED = 201, NT = 38,

with the orthogonal mergeability ledger

PROVEN-MERGEABLE = 349, PROVEN-NOT-MERGEABLE = 242.

The local verifier closes nine evidence families. Observational predecessor coverage is PASS_265; historical exhaustive machine replay remains NT_NOT_EXHAUSTIVE.

Natural next safe boundary. A safe successor may study the 3-local derived triangle and possible self-duality obstruction behind the Cube27 cross-linking; refine the FCC/HCP stacking groupoid by a classifying-space/nerve construction while keeping the Sullivan–Epstein–Vogt codimension-three stratified carrier separate; run the Lyndon–Hochschild–Serre differentials beyond the extension-control H^2 page; resolve the rank-7 PLR fusion scheme under further antipodal/Krein quotients; and strengthen the relative theta/residue descent. On the Kepler branch, R36 remains the active checkpoint until a byte-exact successor is supplied, while acquisition of an exactly compatible Lean runtime is a separate tooling gate. No future Kepler stage or five-branch value is assumed.

References

- [1] D. Sullivan, *A counterexample to the periodic orbit conjecture*, Publications Mathématiques de l’IHÉS **46** (1976), 5–14. Supplied source copy hash-locked in `source_recovery/codim3/`.
- [2] D. B. A. Epstein and E. Vogt, *A counterexample to the Periodic Orbit Conjecture in codimension 3*, Annals of Mathematics **108** (1978), 539–552. Supplied source copy hash-locked in `source_recovery/codim3/`.